3. Permanent magnets are used in microphone etc.

Which of the following statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) Only 2 (d) 1 and 3

296. An electric generator converts
(a) electric energy into sound energy
(b) electric energy into light energy
(c) electric energy into mechanical energy
(d) mechanical energy into electric energy

297. The essential difference between an AC generator and a DC generator has permanent magnet.
(a) AC generator has an electromagnet while a DC generator has permanent magnet
(b) DC generator will generate a higher voltage
(c) AC generator will generate a higher voltage
(d) AC generator has slip rings while DC generator has a commutator

298. The core of a transformer is laminated so that
(a) ratio of the voltages across the secondary and primary is doubled
(b) the weight of the transformer can be kept low
(c) the rusting of the core is prevented
(d) energy loss due to eddy currents is minimised

299. Which one of the following common devices works on the basis of the principle of mutual induction?
(a) Tubelight (b) Transformer
(c) Photodiode (d) LED

300. Consider the following statements
1. Core of a transformer is made up of soft iron.
2. Step-up transformer converts a low voltage of high current into a high voltage of low current.
3. Transformer is used for charging voltage only.

Which of the following statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) None of these

301. Transformer is a kind of appliance that can
1. increase power
2. increase voltage
3. decrease voltage
4. measure current and voltage

Select the correct answer using the codes given below.

(a) 1 and 4 (b) Only 4
(c) 2 and 3 (d) 2, 3 and 4

Previous Years' Questions

302. In step-down transformer, the AC output gives the **2012 II**
(a) current more than the input current
(b) current less than the input current
(c) current equal to the input current
(d) voltage more than the input voltage

303. A rectifier is an electronic device used to convert **2012 II**
(a) AC voltage into DC voltage
(b) DC voltage into AC voltage
(c) sinusoidal pulse into square pulse
(d) None of the above

304. A device which is used in our TV set, computer, radio set for storing the electric charge, is **2012 II**
(a) resistor (b) inductor
(c) capacitor (d) conductor

305. If two conducting spheres are separately charged and then brought in contact **2013 I**
(a) the total energy of the two spheres is conserved
(b) the total charge on the two spheres is conserved
(c) both the total energy and the total charge are conserved
(d) the final potential is always the mean of the original potential of the two spheres

306. When an incandescent electric bulb glows **2014 I**
(a) the electric energy is completely converted into light
(b) the electric energy is partly converted into light energy and partly into heat energy
(c) the light energy is converted into electric energy
(d) The electric energy is converted into magnetic energy

307. A mobile phone charger is **2014 I**
(a) an inverter (b) an UPS
(c) a step-down transformer
(d) a step-up transformer

308. Inactive nitrogen and argon gases are usually used in electric bulbs in order to **2014 II**
(a) increase the intensity of light emitted
(b) increase the life of the filament
(c) make the emitted light coloured
(d) make the production of bulb economical

309. Tungsten is used for the construction of filament in electric bulb because of its **2014 II**
(a) high specific resistance
(b) low specific resistance
(c) high light emitting power
(d) high melting point

310. Electricity is produced through dry cell from **2015 I**
(a) chemical energy
(b) thermal energy
(c) mechanical energy
(d) nuclear energy

311. A wire-bound standard resistor uses manganin or constantan. It is because **2015 I**
(a) these alloys are cheap and easily available
(b) they have high resistivity
(c) they have low resistivity
(d) they have resistivity which almost remains unchanged with temperature

312. Two pieces of conductor of same material and of equal length are connected in series with a cell. One of the two pieces has cross-sectional area double that of the other. Which one of the following statements is correct in this regard? **2015 I**
(a) The thicker one will allow stronger current to pass through it
(b) The thinner one would allow stronger current to pass through it
(c) Same amount of electric current would pass through both the pieces producing more heat in the thicker one
(d) Same amount of electric current would pass through both the pieces producing more heat in the thinner one

313. Which one of the following statements about bar magnet is correct? **2016 I**
(a) The pole strength of the North pole of a bar magnet is larger than that of the South pole.
(b) When a piece of bar magnet is bisected perpendicular to its axis, the North and South poles get separated.
(c) When a piece of bar magnet is bisected perpendicular to its axis, two new bar magnets are formed.
(d) The poles of a bar magnet are unequal in magnitude and opposite in nature.

Modern Physics

314. Which is not true with respect to the cathode rays?
(a) A stream of electrons
(b) Charged particles
(c) Move with speed same as that of light
(d) Can be deflected by magnetic fields

315. Consider the following statements X-rays
1. can pass through aluminium.
2. can be deflected by magnetic field.
3. move with a velocity less than the velocity of ultraviolet ray in vacuum.

Which of the statements given above is/are correct?
(a) 2 and 3 (b) Only 1
(c) 1 and 3 (d) 1 and 2

316. Consider the following statements
1. Soft and hard X-rays differ in frequency as well as velocity.
2. The penetrating power of hard X-rays is more than the penetrating power of soft X-rays.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

317. X-ray will travel minimum distance in
(a) air (b) iron (c) wood (d) water

318. X-ray beam can be deflected by
(a) magnetic field
(b) electric field
(c) Both 'a' and 'b'
(d) None of the above

319. X-rays were discovered by
(a) Johnson (b) Milikan
(c) Roentgen (d) Thomson

320. Positive rays were discovered by
(a) Thomson (b) Goldstein
(c) W Crookes (d) Rutherford

321. Consider the following statements
1. Positive rays are capable of producing physical and chemical changes.
2. Positive rays can produce fluorescence and phosphorescence.
3. Positive rays are not deflected by electric field and magnetic field.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) Only 2

322. Match the following columns.

Column I	Column II
A. Cathode rays	1. Electromagnetic waves
B. X-rays	2. Beam of electron
C. Positive-rays	3. Positively charged particles

Codes

	A	B	C		A	B	C
(a)	3	1	2	(b)	2	1	3
(c)	1	2	3	(d)	1	3	2

323. Consider the following statements
1. Diode laser are used as optical sources in optical communication.
2. Diode laser consume less energy.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

324. An ordinary tubelight used for lighting purpose contains
(a) fluorescent material and an inert gas
(b) one filament, reflective material and mercury vapour
(c) fluorescent material and mercury vapour
(d) two filaments, fluorescent material and mercury vapour

325. Fire fly gives us cold light by virtue of the phenomenon of
(a) fluorescence
(b) phosphorescence
(c) chemiluminescence
(d) effervescence

326. Photoelectric cell is a device of
(a) collecting the photons
(b) measuring the intensity of light
(c) changing the photon energy to mechanical energy
(d) substituting the accumulators by storing the electrical energy

327. What is the difference between a CFL and an LED lamp
1. To produced light, a CFL uses mercury vapour and phosphor while an LED lamp uses semiconductor materials.
2. The average life span of a CFL is much longer than that of an LED lamp.
3. A CFL is less energy efficient as compared to an LED lamp.

Which of the following statements is/are correct?
(a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) All of these

328. Photoelectric cells are used
1. to control the temperature in furnace and in chemical process
2. in automatic switches for street light
3. in transformer
4. in photoelectron sorters

Which of the following statements is/are correct?
(a) 1, 2 and 4 (b) Only 1
(c) Only 2 (d) 1 and 3

329. The majority charge carriers in p-type semiconductors are
(a) electrons (b) protons
(c) holes (d) neutrons

330. If the temperature of a semiconductor is raised, its resistivity will
(a) increase (b) decrease
(c) remains same (d) reduce to zero

331. Consider the following statements
1. Solar cells are used as solar cooker.
2. Solar cells are used in wrist watches, calculators and light meters.
3. Solar cells are used for charging storage batteries.
4. Solar cells are used in artificial satellite

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1, 3 and 4 (d) All of these

332. Laser is a device to produce
(a) a beam of white light
(b) coherent light
(c) microwaves (d) X-rays

333. Which of the following is the correct combination of the inventors and the inventions?
(a) Elisha G Otis — Windmill
(b) Galileo Galilei — Transistor
(c) Sir Frank Whittle — Laser
(d) J L Baird — Television

334. Match the following Columns.

Column I	Column II
A. Ground wave propagation	1. Radio waves received from troposphere
B. Space wave propagation	2. Radio waves travel through atmosphere
C. Sky wave propagation	3. Microwaves cross the ionosphere
D. Satellite communication	4. Radio waves received from ionosphere

Codes

	A	B	C	D
(a)	2	1	4	3
(b)	1	3	4	2
(c)	3	2	1	4
(d)	2	1	3	4

335. The television set receives
(a) magnetic waves
(b) radio waves
(c) microwaves
(d) None of the above

336. Optical fibres are mainly used for which of the following?
(a) Communication
(b) Eye surgery
(c) Weaving
(d) Food industry

Previous Years' Questions

337. Before X-ray examination (coloured X-ray) of the stomach, patients are given suitable salt of barium because **2012 I**
(a) barium salts are white in colour and this helps stomach to appear clearly
(b) barium is a good absorber of X-rays and helps stomach to appear clearly
(c) barium salts are easily available
(d) barium allows X-rays to pass through the stomach

338. Light Emitting Diode (LED) converts **2013 I**
(a) light energy into electrical energy
(b) electrical energy into light energy
(c) thermal energy into light energy
(d) mechanical energy into electrical energy

339. Dual energy X-ray Absorptiometry (DEXA) is used to measure **2013 II**
(a) Spread of solid tumour
(b) bone density
(c) ulcerous growth in stomach
(d) extent of brain haemorrhage

340. X-rays are **2015 II**
(a) deflected by an electric field, but not by a magnetic field
(b) deflected by a magnetic field, but not by an electric field
(c) deflected by both a magnetic field and an electric field
(d) not deflected by an electric field and a magnetic field

ANSWERS

1	b	2	a	3	b	4	b	5	b	6	a	7	b	8	a	9	a	10	a
11	c	12	a	13	c	14	d	15	d	16	c	17	a	18	d	19	c	20	c
21	b	22	d	23	d	24	a	25	b	26	d	27	d	28	d	29	c	30	c
31	c	32	b	33	c	34	a	35	b	36	a	37	d	38	d	39	d	40	a
41	a	42	a	43	a	44	a	45	a	46	c	47	a	48	a	49	a	50	a
51	b	52	b	53	b	54	a	55	a	56	d	57	d	58	c	59	c	60	a
61	d	62	c	63	d	64	a	65	a	66	b	67	a	68	b	69	b	70	b
71	c	72	a	73	d	74	c	75	a	76	d	77	a	78	d	79	d	80	b
81	b	82	a	83	c	84	c	85	a	86	b	87	b	88	a	89	b	90	b
91	b	92	a	93	b	94	c	95	b	96	b	97	b	98	c	99	d	100	a
101	c	102	a	103	c	104	a	105	b	106	c	107	a	108	c	109	c	110	c
111	c	112	d	113	a	114	d	115	b	116	b	117	a	118	a	119	c	120	a
121	c	122	c	123	c	124	d	125	a	126	a	127	c	128	a	129	b	130	a
131	c	132	c	133	a	134	a	135	c	136	a	137	d	138	a	139	a	140	c
141	a	142	d	143	c	144	a	145	d	146	b	147	b	148	a	149	a	150	a
151	a	152	c	153	c	154	c	155	a	156	d	157	a	158	b	159	c	160	a
161	b	162	a	163	a	164	d	165	b	166	c	167	c	168	a	169	d	170	a
171	b	172	b	173	d	174	a	175	a	176	a	177	b	178	d	179	c	180	d
181	b	182	a	183	c	184	c	185	c	186	b	187	b	188	d	189	a	190	c
191	a	192	b	193	a	194	a	195	c	196	b	197	c	198	a	199	c	200	c
201	b	202	d	203	a	204	d	205	c	206	b	207	c	208	d	209	a	210	a
211	c	212	a	213	b	214	a	215	a	216	c	217	a	218	d	219	b	220	c
221	a	222	c	223	a	224	d	225	d	226	b	227	c	228	c	229	b	230	b
231	a	232	b	233	a	234	c	235	b	236	d	237	a	238	a	239	d	240	d
241	c	242	c	243	b	244	a	245	c	246	c	247	d	248	d	249	b	250	b
251	d	252	a	253	c	254	b	255	a	256	d	257	a	258	b	259	c	260	b
261	d	262	a	263	a	264	c	265	a	266	b	267	d	268	c	269	d	270	c
271	c	272	a	273	a	274	a	275	c	276	a	277	a	278	a	279	c	280	b
281	d	282	a	283	a	284	d	285	d	286	a	287	d	288	b	289	d	290	a
291	d	292	c	293	a	294	d	295	d	296	d	297	d	298	d	299	b	300	a
301	c	302	a	303	a	304	c	305	b	306	b	307	c	308	b	309	d	310	a
311	d	312	a	313	c	314	c	315	b	316	c	317	b	318	d	319	c	320	b
321	a	322	b	323	c	324	d	325	c	326	b	327	c	328	a	329	c	330	b
331	d	332	b	333	d	334	a	335	c	336	a	337	b	338	b	339	b	340	d

Solutions

1. (*b*) Given,

$v_1 = 30$ km/h

and $v_2 = 50$ km/h and $v_{av} = ?$

$$v_{av} = \frac{2v_1 v_2}{v_1 + v_2} = \frac{2\times 30 \times 50}{30+50}$$

$$= \frac{3000}{80} = 37.5 \text{ km/h}$$

5. (*b*) From second equation of motion,

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow \quad s = 0 + \frac{1}{2}at^2$$

$$s \propto t^2 \qquad \text{(if } a = \text{constant)}$$

So, object moves with constant or uniform acceleration.

11. (*c*) For maximum cover horizontal range,

$$R_{\max} = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow \quad R_{\max} = \frac{u^2}{g} \qquad (\theta = 45^\circ)$$

18. (*d*) Given, $m = 1$ kg, $F = 1$ N

$F = ma$

$\Rightarrow \quad 1 = 1 \times a$

$\Rightarrow \quad a = 1 \text{ m/s}^2$

36. (*a*) Given,

$m = 2$ kg, $h = 1.5$ m and work done = ?

$g = 9.8 \text{ m/s}^2$

Work done = PE = mgh

$= 2 \times 9.8 \times 1.5 = 29.4$ J

40. (*a*) Let the weight of dog be m kg, then the weight of horse be 10 m kg

$\because \quad \text{KE} = \frac{1}{2}mv^2$ or $K \propto m$

$$\therefore \quad \frac{K_1}{K_2} = \frac{m_1}{m_2} = \frac{m}{10\ m}$$

$K_2 : K_1 = 10 : 1$

41. (*a*) Given, $g = 10 \text{ m/s}^2$, $m = 1$ kg

and $h = 5$ m, PE = ?

$\therefore \quad \text{PE} = mgh = 1 \times 10 \times 5 = 50$ J

44. (*a*) Given,

$g = 9.8 \text{ m/s}^2$,

$m = 200$ kg

PE = 9800 J and $h = ?$

$\because \quad \text{PE} = mgh$

$9800 = 200 \times 9.8 \times h$

$$\Rightarrow \quad h = \frac{9800}{200 \times 9.8} = 5 \text{ m}$$

45. (*a*) Given,

$m_1 = m_2 = m$, $v_1 = 2v$, $v_2 = 3v$

$\because \quad K = \frac{1}{2}mv^2 \quad \Rightarrow \quad K \propto v^2$

$$\therefore \quad \frac{K_1}{K_2} = \left(\frac{v_1}{v_2}\right)^2 = \left(\frac{2v}{3v}\right)^2$$

$\Rightarrow \quad K_1 : K_2 = 4 : 9$

48. (*a*) Time taken by the bus to go from origin to destination with speed of 50 km/h, is

$$t = \frac{300}{50} = 6 \text{ h}$$

$(\because$ distance = 300 km)

Time taken by the bus to go from destination to origin with speed of 60 km/h, is

$$t_2 = \frac{300}{60} = 5 \text{ h}$$

$$\therefore \quad \text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{300 + 300}{6 + 5} = \frac{600}{11}$$

$= 54.55$ km/h

50. (*a*) In this case initial and final position are same. So, displacement will be zero.

Distance travelled by object $= 2\pi r$

$$= 2 \times \frac{22}{7} \times 35$$

$$= 70 \times \frac{22}{7} = 220 \text{ m}$$

53. (*b*) The centripetal force, $F = \dfrac{mv^2}{r}$

Because an electron and a proton are circulating with same speed in circular paths of equal radius

So, $\quad F \propto m$

and $\quad \dfrac{F_e}{F_p} = \dfrac{m_e}{m_p} = \dfrac{m}{2000\ m}$

$F_p = 2000\ F_e$

Therefore, the centripetal force required by the proton is about, 2000 times more than that required by the electron.

55. (*a*) From law of conservation of linear momentum

The total momentum of the system is conserved.

i.e. $\quad m_G v + m_B v_G = 0$

$$\Rightarrow \quad v_G = -\frac{m_B}{m_G} v_G \quad \text{or } v_G \propto \frac{1}{m_G}$$

Therefore, gun is much heavier, so velocity of the recoiling gun is less.

100. (*a*) Stress $= \dfrac{\text{Force}}{\text{Area}} = \dfrac{1 \times 10^4}{\pi \times (2 \times 10^{-3})^2}$

$$= \frac{1 \times 10^4}{12.56 \times 10^{-6}}$$

$= 7.96 \times 10^8 \text{ N/m}^2$

104. (*a*) Radius of the heal, $r = \dfrac{1.0}{2}$

$= 0.5 \text{cm} = 0.005 \text{m}$

Pressure, $P = \dfrac{F}{A} = \dfrac{mg}{\pi r^2}$

$$= \frac{50 \times 9.8}{3.14 \times (0.005)^2}$$

$= 6.2 \times 10^6 \text{ N/m}^2$

107. (*a*) Relative density $= \dfrac{\text{Density of substance}}{\text{Density of water}}$

$$10.8 = \frac{\text{Density of silver}}{1.0 \times 10^3}$$

Density of silver $= 10.8 \times 10^3 \text{ kg/m}^3$

117. (*a*) There are two free surfaces of the film.

So, surface tension, $T = \dfrac{F}{2l}$

$\Rightarrow \quad T \times 2l = F$

$\Rightarrow \quad T \times 2l = Mg$

$$T = \frac{Mg}{2l} = \frac{1.5 \times 10^{-2}}{2 \times (30 \times 10^{-2})}$$

$= 0.025$ N/m

137. (*d*) Given, Celsius, $T_C = -30^\circ$ C, $T_F = ?$

$$\because \quad \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\therefore \quad \frac{-30}{5} = \frac{T_F - 32}{9}$$

$-6 \times 9 = T_F - 32$

$-54 = T_F - 32$

$T_F = 32 - 54$

Fahrenheit $= -22^\circ$ F

139. (*a*) $\quad T_C = -40^\circ$ C

$$\because \quad \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\therefore \quad \frac{-40}{5} = \frac{T_F - 32}{9}$$

$-72 = T_F - 32$

$T_F = 32 - 72$

$T_F = -40^\circ$ F

187. (*b*) Given $l_1 = l$, $l_2 = 4l$ and $T' = ?$

$\because \quad T \propto \sqrt{l}$

$$\frac{T'}{T} = \sqrt{\frac{l_2}{l_1}}$$

$\Rightarrow \quad T' = \sqrt{\dfrac{4l}{l}}\, T$

$\Rightarrow \quad T' = 2T$

197. (*c*) Given, $v = 330$ m/s, $f = 550$ Hz

and $\quad \lambda = ?$

$\because \quad v = f\lambda$

$\therefore \quad 330 = 550 \times \lambda$

$\Rightarrow \quad \lambda = \dfrac{3}{5} = 0.6$ m

201. (*b*) Given, $I = 100\, I_0$

$\because \quad \beta = 10 \log_{10}\left(\dfrac{I}{I_0}\right)$

$\therefore \quad \beta = 10 \log_{10}\left(\dfrac{100\, I_0}{I_0}\right)$

$= 2 \times 10 \log_{10} 10$

$= 20$ dB

Total noise level = (20 dB + 20 dB)

= 40 dB

213. (*b*) According to the formula, $T = 2\pi\sqrt{\dfrac{l}{g}}$

Since, the time period of a simple pendulum does not depend on mass.

Therefore, the time period will be remain same.

220. (*c*) Total distance travelled by the sound in going to the surface and back

= speed × time

So, the distance of the wall from the sound source

$= \dfrac{1}{2} \times$ speed × time

$= \dfrac{1}{2} \times 340 \times 0.3$

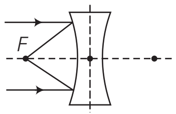
$= 51$ m

232. (*b*) We know that lens formula,

$\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

$\Rightarrow \quad \dfrac{1}{v} + \dfrac{1}{\infty} = \dfrac{1}{f}$

$\Rightarrow \quad v = f$



Clearly, image will be virtual diminished at focus.

233. (*a*) Given, $f = -25$cm (for concave lens)

Power of the lens,

$\because \quad P = \dfrac{1}{f(\text{m})} = \dfrac{100}{f\,(\text{cm})} = \dfrac{100}{25} = 4\text{D}$

So, the nature of lens is concave lens of power 4D.

236. (*d*) Given, the focal length of lens

$f = 80$ cm

The power of the lens,

$P = \dfrac{100}{f\,(\text{cm})} = \dfrac{100}{80} = 1.25$ D

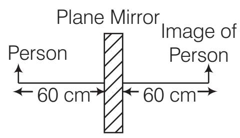
257. (*a*) Given, $f_o = 30$ cm, $f_e = 10$ cm

$\because \quad L = f_o + f_e$

$\therefore \quad L = 30 + 10 = 40$ cm

266. (*b*) The distance between the person and his image in the plane mirror

$= 60 + 60 = 120$ cm $= 1.2$ m



269. (*d*) Myopia is the defect of human eye by virtue of which one cannot see clearly far-off object.

Remedy : using concave lens.

Power of lens $(P) = \dfrac{1}{f}$

Here, $P = -1.25$ D

$\therefore \quad P = \dfrac{1}{f}$

$\Rightarrow \quad f = \dfrac{1}{P} = \dfrac{-1}{1.25} = \dfrac{-4}{5}$ m

$\Rightarrow \quad f = -\dfrac{4}{5} \times 100$ cm $= -80$ cm

or $\quad f = -80$ cm

Since, focal length of the lens is negative, so the lens is concave in nature.

273. (*a*) The resistance of wire

$R = \dfrac{V}{i}$

$\Rightarrow \quad R = \dfrac{0.1}{2} = 0.05\,\Omega$

281. (*d*) According to the question,

$R_1 + R_2 = 9 \qquad \ldots\text{(i)}$

and $\quad \dfrac{R_1 + R_2}{R_1 R_2} = \dfrac{1}{2}$

$\Rightarrow \quad \dfrac{R_1 R_2}{R_1 + R_2} = 2$

$\Rightarrow \quad R_1 R_2 = 18 \qquad \ldots\text{(ii)}$

We have,

$(R_1 - R_2)^2 = (R_1 + R_2)^2 - 4R_1R_2$

$= (9)^2 - 4 \times 18 = 81 - 72$

$(R_1 - R_2)^2 = 9$

$\Rightarrow \quad R_1 - R_2 = 3 \qquad \ldots\text{(iii)}$

On solving Eqs. (i) and (iii), we get

$R_1 = 6\ \Omega$ and $R_2 = 3\ \Omega$

284. (*d*) Power of electric bulb, $P = \dfrac{V^2}{R}$

$\therefore$ So, the resistance of bulb, $R = \dfrac{V^2}{P}$

$R = \dfrac{240 \times 240}{60}$

$\Rightarrow \quad R = 960\ \Omega$

292. (*c*) The effective resistance

$\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$

$\Rightarrow \quad \dfrac{1}{R} = \dfrac{1}{3} + \dfrac{1}{3} + \dfrac{1}{3}$

$\dfrac{1}{R} = \dfrac{1+1+1}{3}$

$\Rightarrow \quad \dfrac{1}{R} = \dfrac{3}{3}$

$\Rightarrow \quad R = 1\,\Omega$

312. (*a*) In series, the current flows through each conductor is same.

Heat produced by resistance R, $H = i^2Rt$

Resistance of thinner wire, $R_1 = \rho\dfrac{l}{A}$

Resistance of thicker wire,

$R_2 = \rho\dfrac{l}{2A} \quad \Rightarrow \quad R_2 = \dfrac{R_1}{2}$

$\therefore$ Heat produced in the thinner wire,

$H_1 = i^2 R_1 t$

Heat produced in the thicker wire,

$H_2 = i^2 R_2 t$

$H_2 = \dfrac{i^2 R_1 t}{2}$

$= \dfrac{H_1}{2}$

So, same amount of electric current would pass through both the pieces producing more heat in the thinner one.